

Ian McCallum

FOUNDER & LEAD ENGINEER · BEAT THE CLOCK

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Naperville, Illinois

Founder and lead engineer of **Beat the Clock**, where I designed and built Clockwork, an AI operations platform for trades businesses that turns a missed call into a drafted reply, a priced quote, a confirmed appointment, and a collected invoice. I work in Python, prompt engineering, and applied AI systems design, with a bias toward software that refuses to guess when the answer costs somebody money. Incoming Finance and Data Science student at the University of Illinois' Gies College of Business.

EXPERIENCE

Beat the Clock

Dec 2024 – Present

Founder & Lead Engineer

- Built **Clockwork**, an AI operations platform for trades businesses, as sole architect and engineer: ten automated stages across eight screens, from catching a missed call to planning the next day's schedule overnight.
- Designed the product around an **approval-gated agent**. Every message is drafted for the owner to approve, and the agent earns permission to send a message type on its own only after enough drafts pass unedited, revocable in one tap.
- Separated **deterministic pricing from model output** so quotes compute from each customer's own rate card, eliminating a class of AI failure where a hallucinated number costs real money.
- Wrote a ten-rule voice specification for the agent and enforced it in the test suite; any violation fails the build. Architected multi-tenant support with strict data isolation between locations.

Luminate · Virtual Enterprise International

2025 – 2026

Lead Developer

1st Place, National · 1st Place, Midwest Regional
E-Commerce Website, Virtual Enterprise International

- Built the full storefront for a glow-in-the-dark sports equipment brand: product catalog, curated bundles, and cart across training gear and party lines.
- Designed and coded an **in-site arcade** with three playable games that award points redeemable for promo codes, giving customers a reason to return between purchases.

Freelance Web Development

2023 – Present

Web Developer

- Shipped custom sites for paying clients including **Imanol VillaGomez** (concert videographer for Charli XCX and Playboi Carti), **Fortivus Academy** (the largest rugby program in Naperville), TMM Photography, and Nics Marketing.
- Built beatyourclock.com end to end, including an interactive product simulation and a free assessment tool that quantifies the hours a shop loses each week.

Vokel

2019 – 2021

Creator, YouTube & Twitch

- Grew a gaming channel to nearly 500 subscribers and reached Twitch Affiliate status, editing every video in Premiere Pro and After Effects.

EDUCATION

University of Illinois Urbana-Champaign

Gies College of Business · Expected 2030
Finance & Data Science

Metee Valley High School

Aurora, IL · Class of 2026 · GPA 4.25

8 AP courses including Statistics, Macroeconomics, and U.S. Government & Politics. Virtual Enterprise International.

SKILLS

AI systems design

Prompt engineering

LLM integration

Python

JavaScript & React

Product design

Client communication

Self-directed

Autodidactic

TECHNICAL

Languages: Python, JavaScript, HTML, CSS, React

AI & Systems: Agent design, human-in-the-loop approval workflows, automated testing of model output, multi-tenant architecture

Integrations: Messaging/SMS, payments, scheduling, invoicing

AWARDS

1st Place, National · E-Commerce Website, VEI

1st Place, Midwest Regional · VEI

Illinois State Scholar

AP Scholar with Distinction

Magna Cum Laude · Seal of Biliteracy

ALSO

Target · Food Dept. · 2024–Present

Coldwell Banker Dan Firks · Real Estate Runner
· Summer 2025

Youth Soccer Referee · 2021–2024

UNICEF Club · Board Member

National Honor Society